

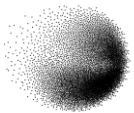


DÉPART
DE SENTIER

BrightCon 2026
Intermediate course day

Time-explicit LCA with **bw_timex**

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PSI



RWTH AACHEN
UNIVERSITY



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Leiden
The Netherlands



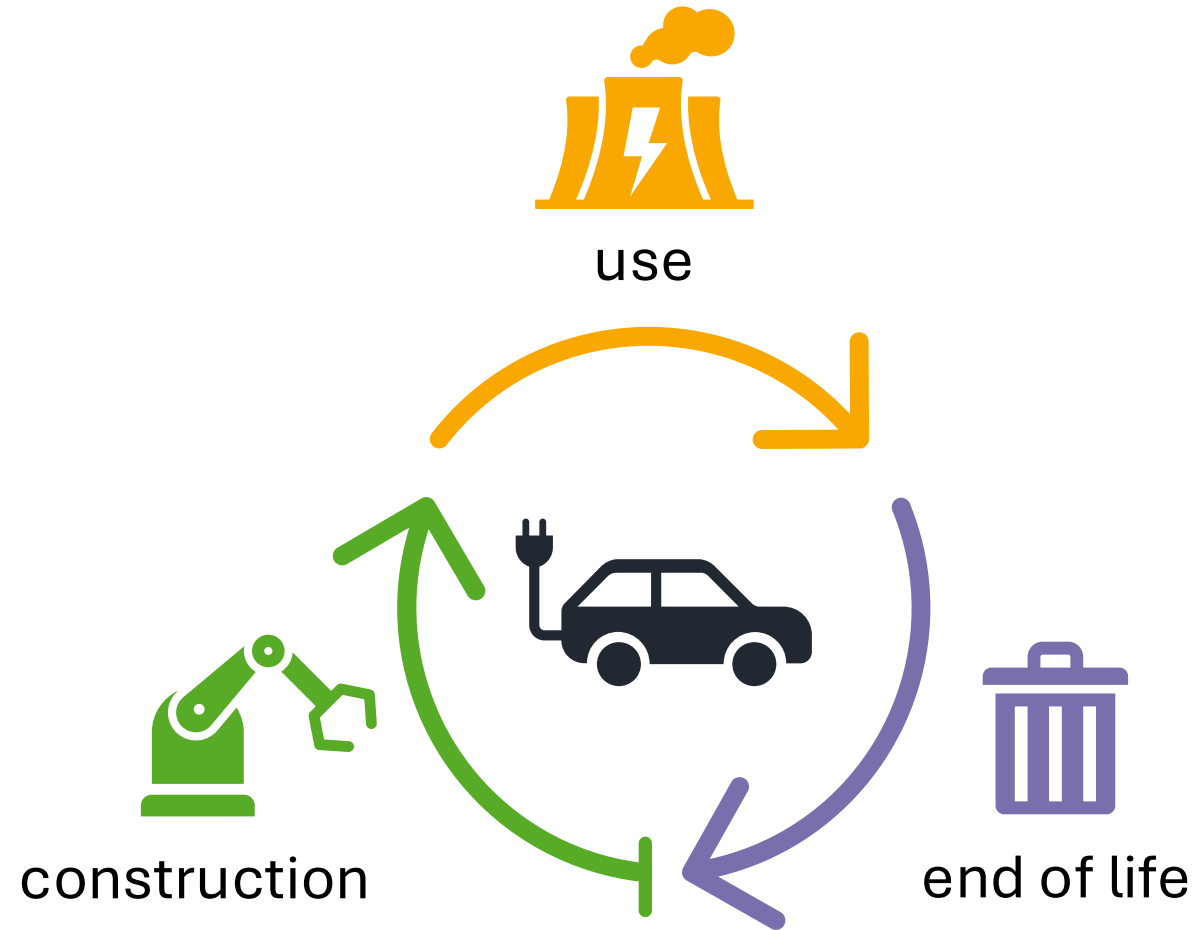
Course day schedule

Time	Activity
09:00 - 12:00	Timex (break included)
12:00 - 13:00	Lunch Break
13:00 - 16:00	Systems Dynamics & openLCA (break included)
16:00 - 17:00	Discussion and Reflections

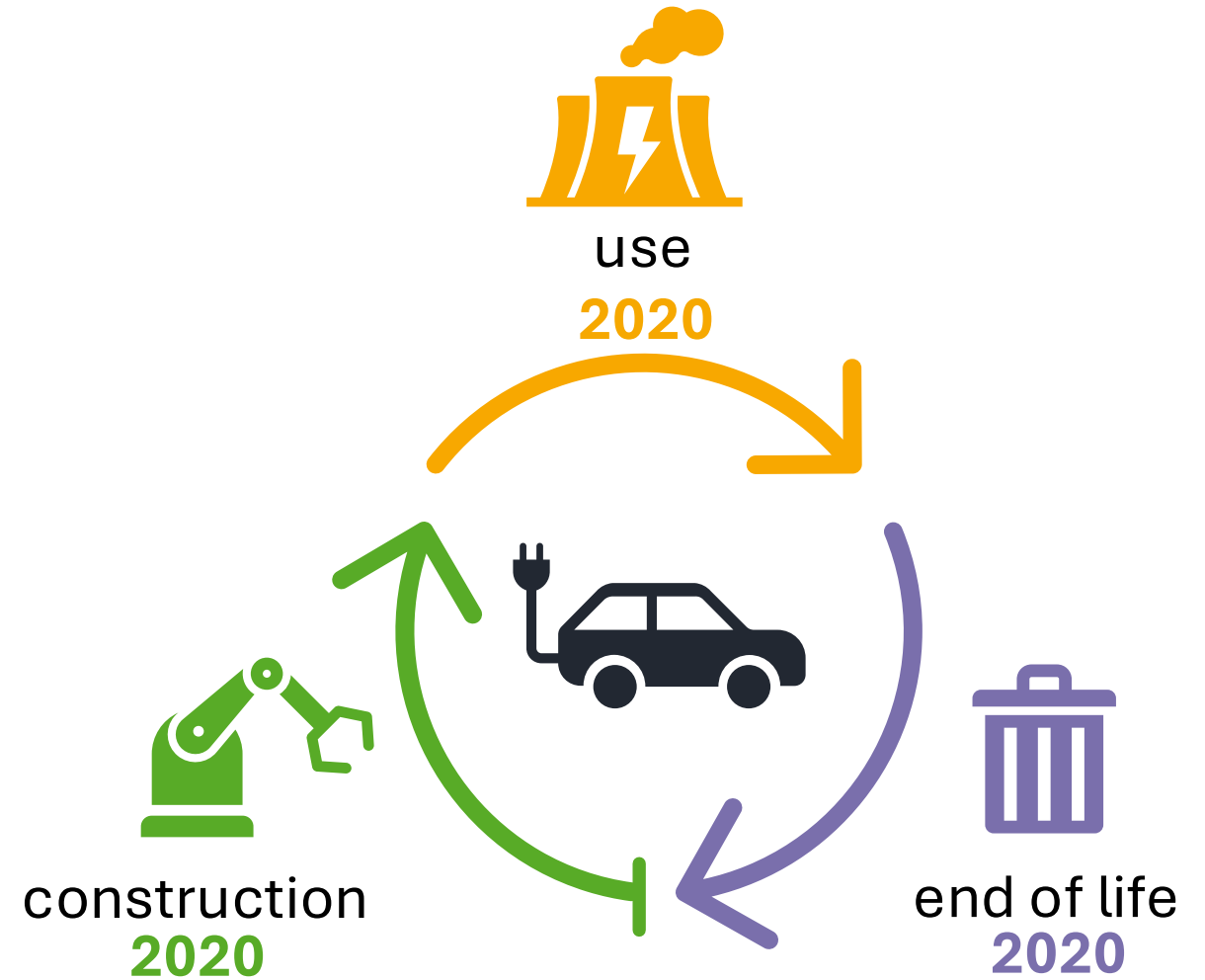
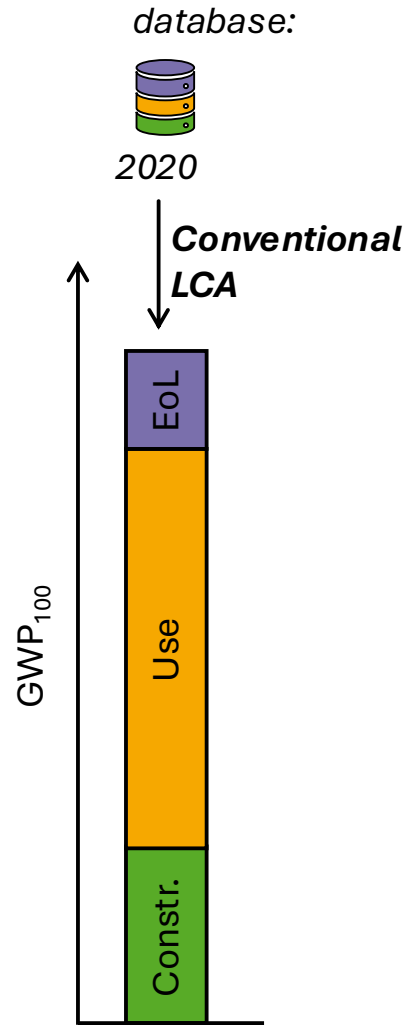
Morning block schedule

Time	Activity
09:00 – 09:15	Welcome and short introduction round
09:15 – 09:30	Time-explicit LCA in a nutshell + walkthrough
09:30 – 10:30	Interactive time-explicit case study
10:30 – 10:50	-- Break --
10:50 – 11:40	Interactive dynamic impact assessment
11:40 – 12:00	Discussion

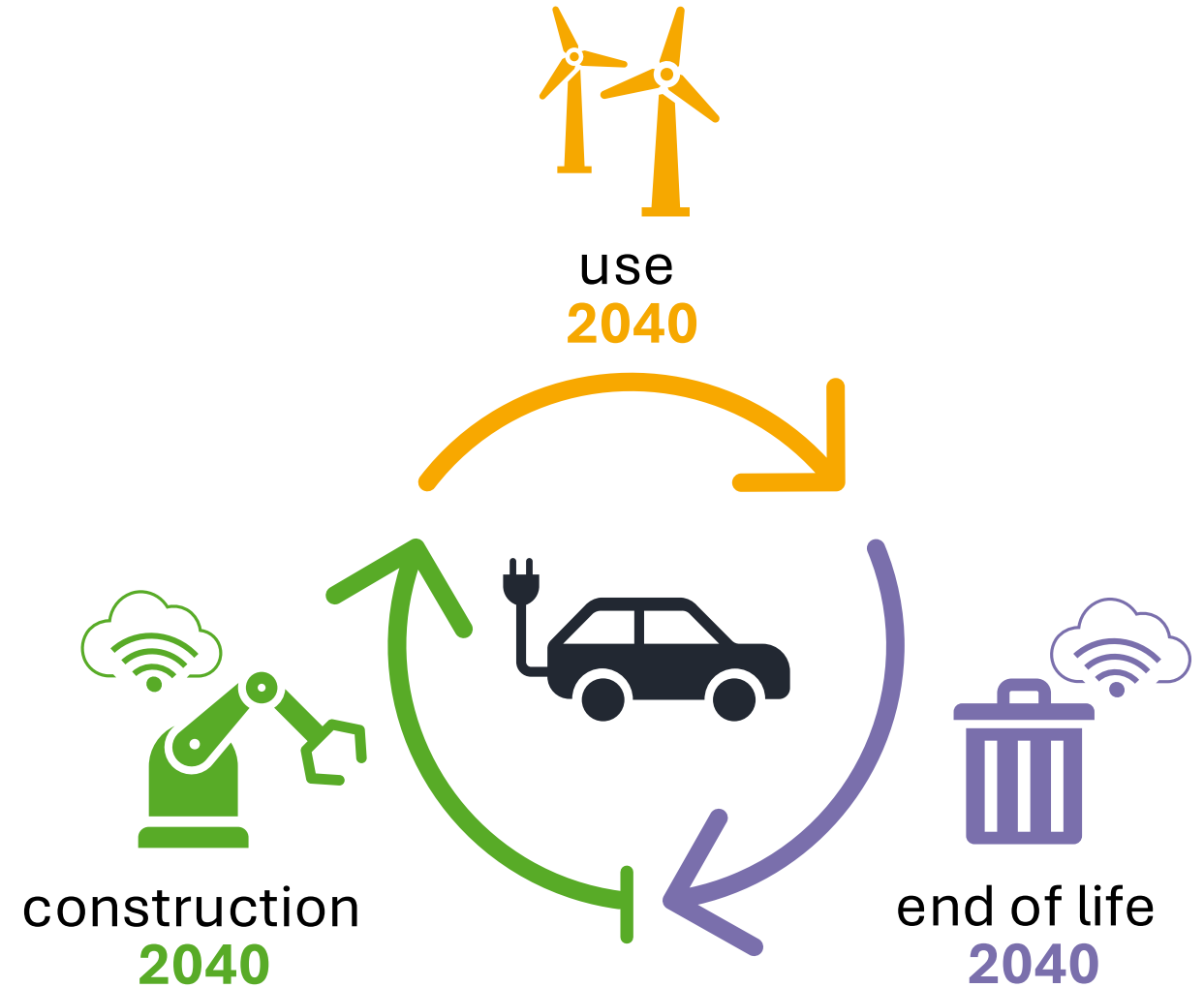
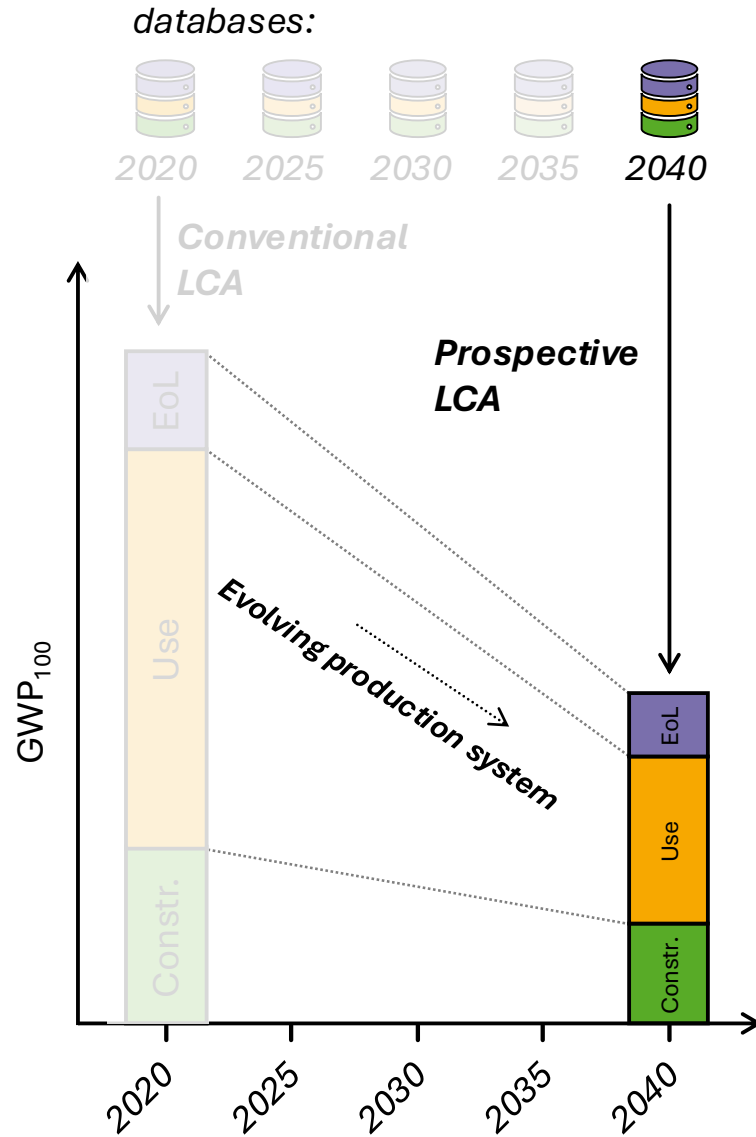
Life cycle of an electric vehicle



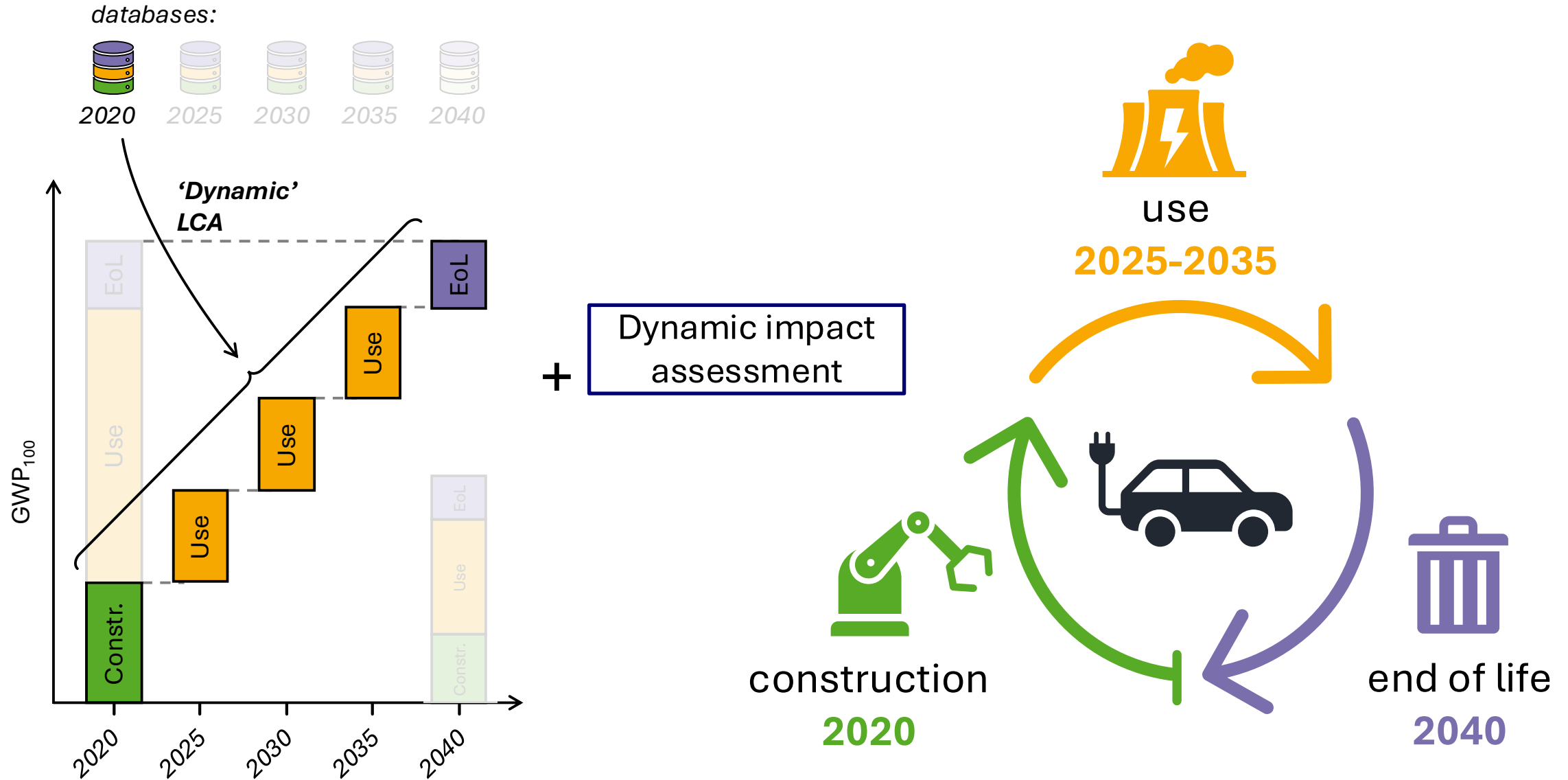
‘Conventional’ LCA: “system today”



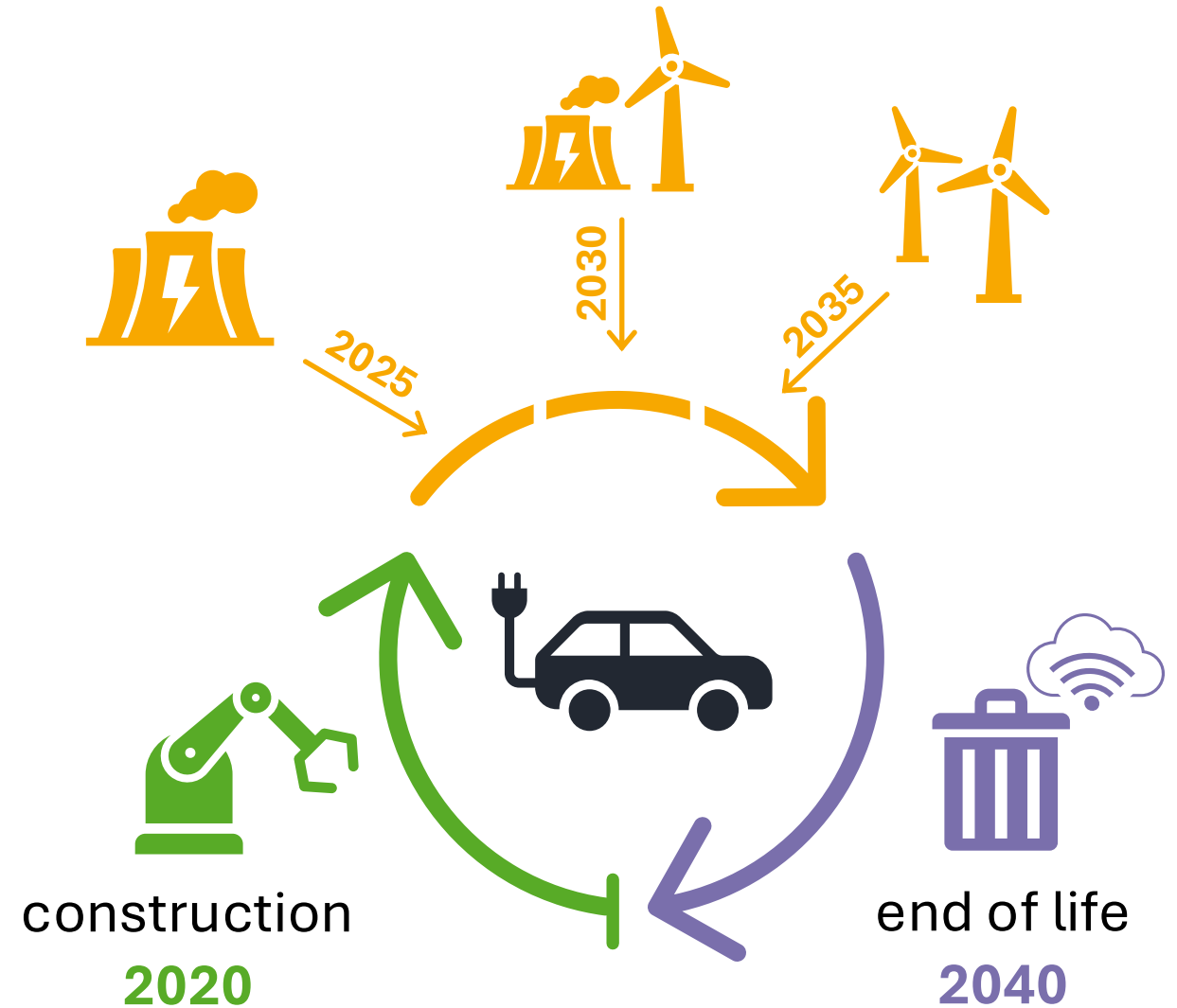
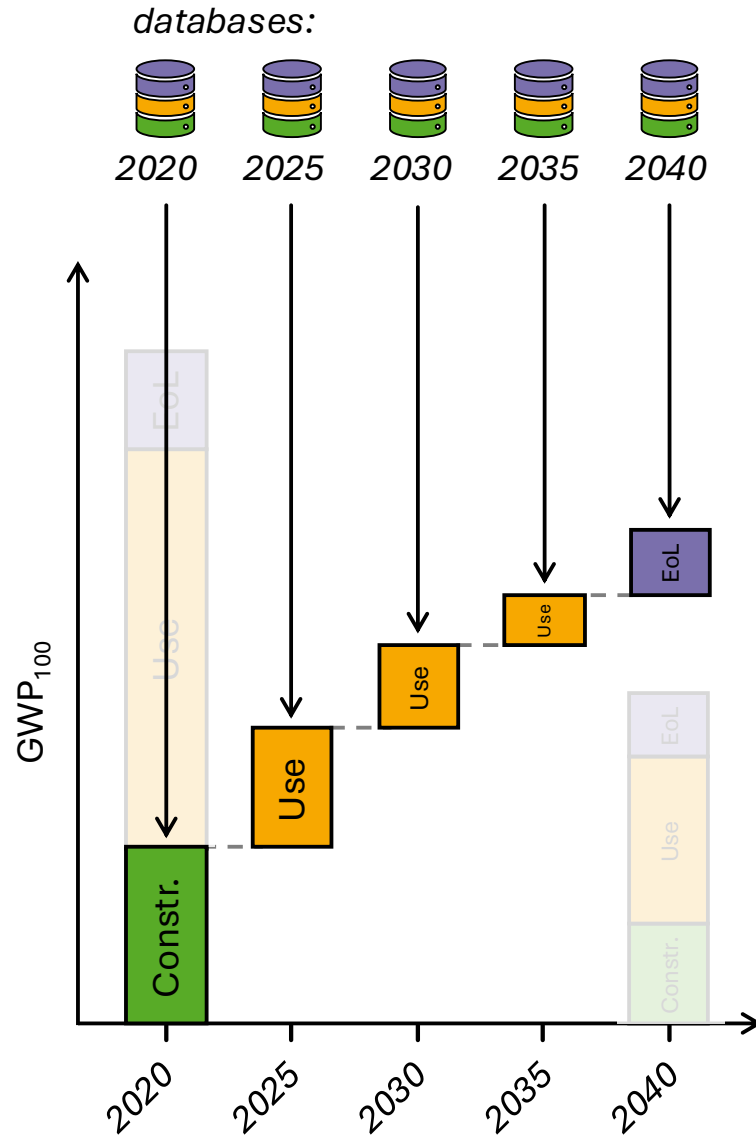
Temporal Evolution: “prospective state of the system”



Temporal distribution: “the system over time”



Time-explicit LCA: “Temporal distribution + evolution”



Summary

Table 1 Schematic overview of how different LCA methods typically treat temporal distribution and temporal evolution at the life cycle inventory (LCI) and life cycle impact assessment (LCIA) phase

	Life cycle inventory		Life cycle impact assessment	
	Temporal distribution	Temporal evolution	Temporal distribution	Temporal evolution
Conventional LCA	No, 1 current timestep	No	No	No
Dynamic LCA	Yes, multiple timesteps	Rarely	Yes	No
Prospective LCA	No, 1 future timestep	Yes	No	Rarely
Retrospective LCA	No, 1 past timestep	Yes	No	No
Time-explicit LCA (<i>proposed in this study</i>)	Yes, multiple timesteps	Yes	Yes *	Yes*

* optional



In a time-explicit LCA, you model a system *over time*, with time-specific processes and amounts, and *time-dependent* impact assessment of emissions.

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